Lab experiment

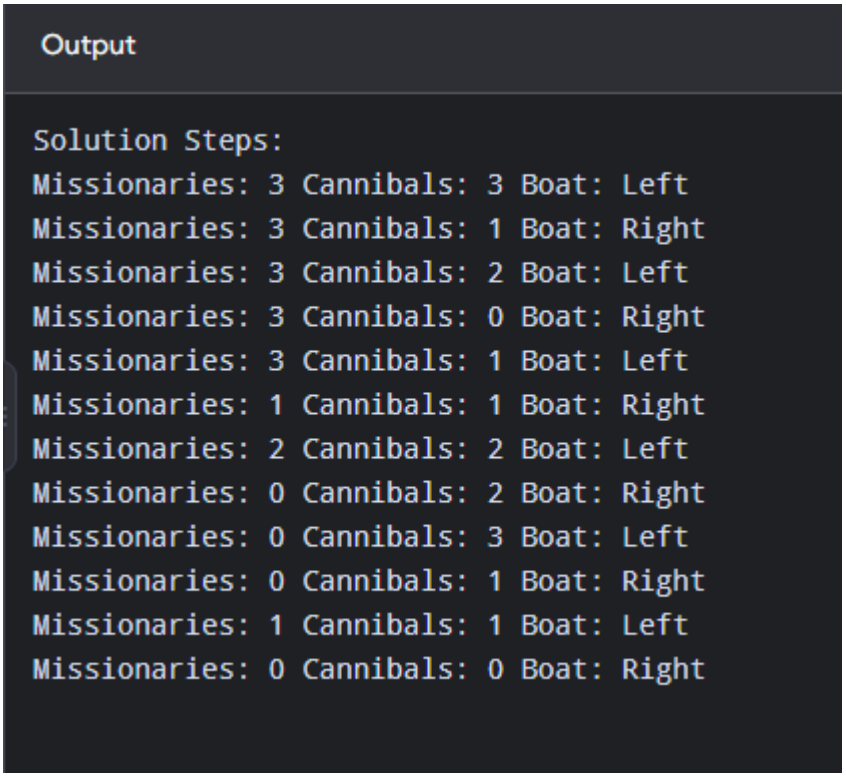
Missionaries and Cannibals Problem

coding:

```
from collections import deque

# State: (Missionaries, Cannibals, Boat)
start = (3, 3, 0)
goal = (0, 0, 1)

# Possible boat movements
moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]

# Check whether a state is valid
def is_valid(state):
    m, c, b = state

    mr = 3 - m
    cr = 3 - c

    if m < 0 or c < 0 or m > 3 or c > 3:
        return False

    if m > 0 and m < c:
        return False

    if mr > 0 and mr < cr:
        return False

    return True

# BFS algorithm
def solve():
    queue = deque([start])
    visited = {start}
    parent = {start: None}

    while queue:
        current = queue.popleft()

        if current == goal:
            break
```

```
        m, c, boat = current

        for dm, dc in moves:
            if boat == 0:
                next_state = (m - dm, c - dc, 1)
            else:
                next_state = (m + dm, c + dc, 0)

            if is_valid(next_state) and next_state not in visited:
                visited.add(next_state)
                parent[next_state] = current
                queue.append(next_state)

    # Reconstruct solution path
    if goal not in parent:
        print("No solution found")
        return

    path = []
    state = goal

    while state is not None:
        path.append(state)
        state = parent[state]

    path.reverse()

    print("Solution Steps:")

    for step in path:
        m, c, boat = step

        if boat == 0:
            side = "Left"
        else:
            side = "Right"

        print("Missionaries:", m,
              "Cannibals:", c,
              "Boat:", side)

# Main program
solve()
```

Output:


Output
Solution Steps:
Missionaries: 3 Cannibals: 3 Boat: Left
Missionaries: 3 Cannibals: 1 Boat: Right
Missionaries: 3 Cannibals: 2 Boat: Left
Missionaries: 3 Cannibals: 0 Boat: Right
Missionaries: 3 Cannibals: 1 Boat: Left
Missionaries: 1 Cannibals: 1 Boat: Right
Missionaries: 2 Cannibals: 2 Boat: Left
Missionaries: 0 Cannibals: 2 Boat: Right
Missionaries: 0 Cannibals: 3 Boat: Left
Missionaries: 0 Cannibals: 1 Boat: Right
Missionaries: 1 Cannibals: 1 Boat: Left
Missionaries: 0 Cannibals: 0 Boat: Right